Finally, if a block is provided, both `create` and `new` will yield the new object to that block for initialization:

```
user = User.new do |u|
  u.name = "David"
  u.occupation = "Code Artist"
end
```

5.2 Read

Active Record provides a rich API for accessing data within a database. Below are a few examples of different data access methods provided by Active Record.

```
# return a collection with all users
users = User.all
# return the first user
user = User.first
# return the first user named David
david = User.find_by(name: 'David')
# find all users named David who are Code Artists and sort by
created_at in reverse chronological order
users = User.where(name: 'David', occupation: 'Code
Artist').order('created_at DESC')
```

You can learn more about querying an Active Record model in the Active Record Query Interfaceguide.

5.3 Update

Once an Active Record object has been retrieved, its attributes can be modified and it can be saved to the database.

```
user = User.find_by(name: 'David')
user.name = 'Dave'
user.save
```

A shorthand for this is to use a hash mapping attribute names to the desired value, like so:

```
user = User.find_by(name: 'David')
user.update(name: 'Dave')
```

This is most useful when updating several attributes at once. If, on the other hand, you'd like to update several records in bulk, you may find the `update_all` class method useful:

```
User.update_all "max_login_attempts = 3, must_change_password =
'true'"
```

5.4 Delete

Likewise, once retrieved an Active Record object can be destroyed which removes it from the database.

```
user = User.find_by(name: 'David')
user.destroy
```

6 Validations

Active Record allows you to validate the state of a model before it gets written into the database. There are several methods that you can use to check your models and validate that an attribute value is not empty, is unique and not already in the database, follows a specific format and many more.

Validation is a very important issue to consider when persisting to the database, so the methods `save`and `update` take it into account when running: they return `false` when validation fails and they didn't actually perform any operation on the database. All of these have a bang counterpart (that